Task 3: Visualization and Interpretation

1. Objective

The objective of this task is to visualize biological data using Python and interpret the results scientifically.

2. Dataset

An illustrative dataset containing expression values for five samples was used to demonstrate biological data visualization. The values are hypothetical and do not represent actual experimental measurements.

3. Tools Used

- Python
- Google Colab
- Matplotlib library

4. Results and Observations

A bar chart was created to compare the illustrative expression values across five samples.

Sample 4 has the highest value of 25, while Sample 1 has the lowest value of 12. Sample 2 has a value of 18, Sample 3 has a value of 15, and Sample 5 has a value of 20.

5. Scientific Interpretation

The bar chart provides a visual comparison of values across the five samples. It demonstrates how data visualization can help identify differences and patterns in a dataset.

6. Limitations: What the Chart Does Not Prove

The dataset contains hypothetical values rather than real biological measurements. Therefore, the chart does not establish actual differences in gene expression, statistical significance, or causal relationships. Further analysis using validated experimental data and appropriate statistical methods would be required to draw biological conclusions.

7. Conclusion

This task demonstrates how Python and Matplotlib can be used to visualize and interpret biological data. It also highlights the importance of distinguishing illustrative patterns from

scientifically validated findings.

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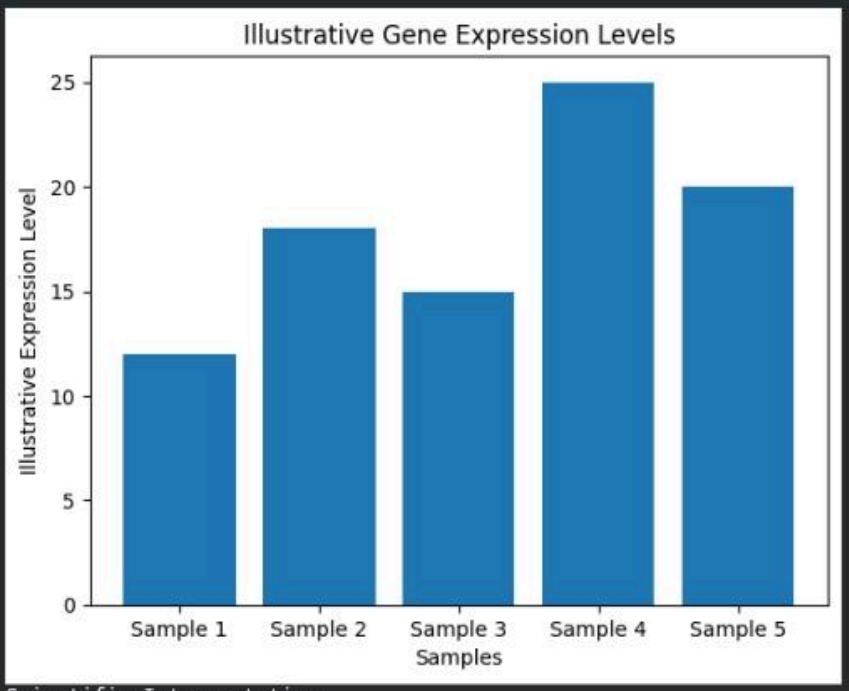
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```
import matplotlib.pyplot as plt

samples = ["Sample 1", "Sample 2", "Sample 3", "Sample 4", "Sample 5"]
values = [12, 18, 15, 25, 20]

plt.bar(samples, values)
plt.xlabel("Samples")
plt.ylabel("Illustrative Expression Level")
plt.title("Illustrative Gene Expression Levels")
plt.show()

print("Scientific Interpretation")
print("Sample 4 has the highest value.")
print("Sample 1 has the lowest value.")
print("The data are illustrative, not real experimental measurements.")
print("The chart does not prove causation or statistical significance.")
```




```
Scientific Interpretation
Sample 4 has the highest value.
Sample 1 has the lowest value.
The data are illustrative, not real experimental measurements.
The chart does not prove causation or statistical significance.
```